Through-Wall Human Activity Recognition With Complex-Valued Range–Time–Doppler Feature and Region-Vectorization ConvGRU

Longzhen Tang, Shisheng Guo, *Member, IEEE*, Qiang Jian, Guolong Cui, *Senior Member, IEEE*, Lingjiang Kong, *Senior Member, IEEE*, and Xiaobo Yang

***Abstract*— In this article, we consider a high-accuracy and low-complexity method for recognizing human activities behind wall. As the amount of information conveyed by data representation directly affects the recognition accuracy of the network, we construct the 3-D complex-valued feature for human activity recognition (HAR). In light of high network complexity introduced by 3-D complex-valued data, we devise a low time and space complexity network named convolutional gated recurrent unit based on region-vectorization (RV-ConvGRU). Keystone transform is utilized to process the radar echo and generate 3-D complex-valued range-time-Doppler (RTD) data first, which provides high-frequency resolution and abundant feature information. Then, the real and imaginary parts of the complex-valued RTD are separately fed into a feature extraction module to comprehensively extract their respective features. Specifically, the real or imaginary part of the RTD is divided into multiple regions, which are then converted into regional vectors and reordered as channels to reduce the time and space complexity of the subsequent network. The reconfigured features are then input into the convolutional gated recurrent unit (ConvGRU) to extract global and temporal features, with the channel attention mechanism for feature selecting. The features of the real and imaginary parts are fused and then classified by the classifier finally. The experiments verify that the proposed method is effective, achieving the highest recognition accuracy of 99.23% with an input sequence of 1.44 s.**

***Index Terms*— Convolutional gated recurrent unit (ConvGRU), human activity recognition (HAR), range–time–Doppler (RTD) feature, through-wall radar.**

I. Introduction

HUMAN activity recognition (HAR) is extensively applied in various fields, such as security monitoring [1], traffic safety [2], and healthcare [3], [4]. Generally, human activity sequences are collected through sensors. The visual-based approach, which captures intuitive image recordings of human activity, has been widely used. However, the effectiveness of camera is inevitably limited by various constraints, including occlusion, reliance on lighting condition, and the potential privacy infringement. As a consequence, many researchers focus on exploring and researching nonvisual sensor-based HAR. While wearable sensor is hindered by limited battery life and cumbersome usage, radar, functioning as an ambient sensor, effectively circumvents these limitations and presents superior range resolution and antiinterference capability. Furthermore, through-wall radar provides a greater variety of application scenarios, such as urban warfare and antiterrorism operations [5], [6].

Typically, radar echoes of human activities are transformed into data representations that contain features related to human activity through specific data analysis techniques. Subsequently, appropriate methods are applied to extract the human activity features from these data representations, which are then utilized to accomplish classification tasks [7], [8].

As radar echoes carry abundant velocity information of the scattering points and time-varying distance information of human movement, they are commonly processed as time-Doppler (TD) frequency, range-time (RT), and range-Doppler (RD) frequency images [9], [10], [11]. In [12], six distinct features are manually extracted from the TD image, which are subsequently fed into a support vector machine (SVM) to classify seven human behaviors, and the feasibility of recognition under attenuation caused by wall is verified. Qi et al. [13] extract eight features from TD image and construct position-information-indexed classifier to deal with the confusion from through-wall attenuation. To tackle the issue of weak Doppler features in through-wall scenario, a novel time-frequency transform is proposed to generate TD image, and eight interpretable features are extracted from the TD image for recognition [14]. Due to the rapid advancements in deep learning technology, neural network with the capacity of automatic feature extraction is increasingly utilized in HAR to address issues of researcher knowledge dependence and task-specific limitations inherent in manual feature extraction [7]. Convolutional neural network (CNN) is endowed with remarkable ability in extracting spatial features, rendering it as automatic feature extractor in HAR [15], [16], [17]. Seyfioglu et al. [18] conduct an analysis on the effectiveness of micro-Doppler signatures and put forward deep

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convolutional autoencoder for obtaining initial parameters of feature extractor under unsupervised training. Additionally, in order to fully exploit the temporal information contained in human behavioral sequence, recurrent neural network (RNN) and its variants are also considered as a part of the feature extractor [19], [20], [21]. ResNet-18 and long short-term memory (LSTM) are employed on RD map to identify normal, autonomous, and distracted driving behaviors [22]. In [23], a through-wall HAR method is proposed, where the autoencoder network is used for dimensionality reduction and unsupervised feature learning on RT image, and bidirectional gated recurrent units (BiGRUs) are introduced to effectively capture the sequential features.

Due to the apparent impact of feature richness on recognition performance, many researchers explore the integration of multiple information to enhance recognition accuracy, rather than relying solely on single 2-D feature images [24]. In [25], to obtain overall representation of human activity behind wall, the researchers propose a multidomain fusion method with TD, RD, and RT images. Jia et al. [26] improve the recognition accuracy by integrating information from the RT and TD images of multiple frequency points in the stepped frequency continuous wave (SFCW) radar. Furthermore, the 3-D range–time–Doppler (RTD) data composed of information from RT, RD, and TD images is gradually utilized for characterizing human activity [27], [28]. Wang et al. [29] project 3-D RTD orthogonally onto TD, RT, and RD planes to create data representations, and build a three-branch EfficientNet to accurately classify activities. Kim and Seo [30] generate wavelet-based TD spectrograms along range units and construct a multibranch range-based CNN according to the range units traversed by the target motion. The 3-D RTD data are converted into point clouds and then recognized and classified through PointNet [31].

However, these works based on 3-D RTD data only utilize amplitude information while ignore phase information, despite the effectiveness of which has been demonstrated in some studies [32], [33], [34]. Guo et al. [35] employ the magnitude RDT point-cloud and the RD phase image as inputs of the neural network. Nevertheless, the approach solely utilizing the RD phase image overlooks the temporal correlation present in the phase information. Furthermore, the attenuation in through-wall scenario weakens the available Doppler information [36], [37]. Therefore, it is essential to obtain more complete and accurate information to achieve high-precision recognition under through-wall scenario. In this article, we propose an HAR method using through-wall radar with high accuracy and low complexity based on complex-valued 3-D RTD data. The form of complex-valued RTD data effectively preserves the complete information of the real part, imaginary part, range, time, Doppler, and the interdependence among them. To begin with, we utilize the keystone transform, which improves Doppler frequency resolution and mitigates the Doppler broadening effect [38], to process radar echoes and generate complex-valued RTD data. Subsequently, a region-vectorization-based model is designed to address the issue of the surge in computational complexity resulting from complex-valued 3-D data. Moreover, to fully exploit the real and imaginary components of the complex-valued data, a dual-channel feature extraction module is designed to separately process the real and imaginary parts of the RTD, and feature-level fusion strategy is adopted for classification. Specifically, the real or imaginary value of RTD data is divided into multiple regional blocks, and each regional block is transformed into a feature vector that retains local features of the region. Next, the regional feature vectors are rearranged to form the channels of each corresponding region, and the channel attention mechanism is introduced to enhance network's selectivity toward local features. Convolutional-gated recurrent unit (ConvGRU) is utilized to capture the global spatial and temporal characteristics of the processed 3-D RTD data. Ultimately, the extracted real and imaginary features are concatenated for fusion and sent to multilayer perceptrons (MLPs) for classification.

The contributions of this article are summarized as follows.

1) The proposed HAR method takes complex-valued 3-D RTD as data representation to enrich human behavior information, and applies the keystone transform to obtain more accurate Doppler frequency information.
2) The region-vectorization module is designed to solve the computational burden brought about by 3-D complex-valued data. By segmenting 3-D data into blocks, converting blocks into vectors, and reordering vectors into channels, the computations within each channel are transformed into interchannel computations to diminish the size of output features, significantly reducing the time and spatial complexity of subsequent network.
3) The proposed network can make full use of the information contained in the complex-valued RTD data by considering various feature dimensions. In detail, the region-vectorization model and convolution operations in the ConvGRU are responsible for capturing local and global spatial features of the RD domain. Additionally, the GRU structure in the ConvGRU module connects the temporal features between RD maps at different times. Finally, the dual-channel architecture enables comprehensive utilization of both the real and imaginary parts.
4) Extensive experimental evaluations are conducted to validate the effectiveness of the proposed HAR approach.

The subsequent content of this article is organized as follows. Section II models the echo from through-wall human activity and describes the generation method of complex-valued 3-D RTD. In Section III, we present the complete approach of HAR and elaborate on each module of the network. Then, datasets and implementation details of experiments are provided in Section IV. Experimental results are demonstrated and analyzed in Section V. Finally, Section VI concludes this article.

II. Signal Model and Feature Expression Generation

A. Radar Echo Model

In this study, an ultrawideband SFCW radar system is employed to detect human target behind wall, as shown in Fig. 1. The ultrawideband SFCW radar transmits multiple

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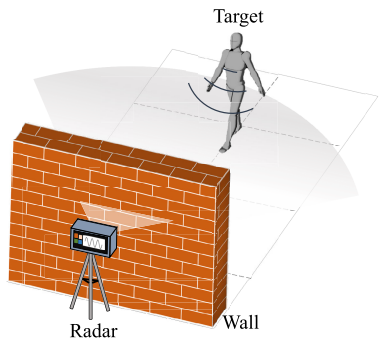

Fig. 1. Scenario for through-wall radar detecting human motion.

single-frequency signals with an initial frequency of f_0, an incremental frequency step of Δf, and a time step of Δt [39], which can be mathematically expressed as

$$S_t(t) = \sum_{m=1}^{M} e^{j2\pi f_m t} \operatorname{rect}\left(\frac{t - m\Delta t + \frac{\Delta t}{2}}{\Delta t}\right) \tag{1}$$

where $f_m = f_0 + (m-1)\Delta f$, and rect$(\cdot)$ denotes rectangular window function, defined as

$$\operatorname{rect}(t) = \begin{cases} 1, & |t| \le \dfrac{1}{2} \\ 0, & |t| > \dfrac{1}{2}. \end{cases} \tag{2}$$

Transmitted electromagnetic wave is reflected back upon encountering the human target, and then is received by the radar antenna. The human target is modeled as a combination of G scattering points, so the radar echoes can be represented as

$$S_r(t) = \sum_{g=1}^{G}\sum_{m=1}^{M} \sigma_g e^{j2\pi f_m (t-\tau_g(t))} \times \operatorname{rect}\left(\frac{t - \tau_g(t) - m\Delta t + \frac{\Delta t}{2}}{\Delta t}\right) + S_w(t) + S_n(t) \tag{3}$$

where σ_g is the scattering coefficient of the gth scattering point and $S_w(t)$ indicates strong reflected echoes from the wall. $S_n(t)$ represents the ensemble of environmental clutter, coupling waves between transmitting and receiving antennas, and noise. Besides, $\tau_g(t)$ denotes the time required for transmitted wave to reach the gth scattering point and return to the radar receiver, which is given by

$$\tau_g(t) = \frac{2\left(R_0 + vt + \sqrt{\epsilon_w} R_w\right)}{c} \tag{4}$$

where R_0 is initial distance between the gth target scattering point and radar, v represents the velocity of scattering point, ϵ_w denotes the dielectric constant of wall, and R_w is the distance traveled by the electromagnetic wave within the wall.

By performing demodulation, sampling, and clutter suppression on the collected radar echoes, and combining N cycles of received signals together, a data matrix $\mathbf{S}$ with size of $N \times M$ can be obtained, where each row of the matrix corresponds to a single-cycle signal, and each column corresponds to a single-frequency point. In other words, the matrix $\mathbf{S}$ can be viewed as N cycles of echoes, with each cycle containing M frequency points. The received echo data located at the nth row and mth column of the matrix $\mathbf{S}$ can be written as

$$S_r(n, m) = \sum_{g=1}^{G} \sigma_g e^{-j2\pi f_m \tau_g(t_m)} \tag{5}$$

where $t_m = (n-1)T + (m-1)\Delta t$.

B. Generation of Complex-Valued RTD

The traditional method obtains the fast time/slow-time domain by directly performing inverse fast Fourier transform (IFFT) on the rows of the data matrix $\mathbf{S}$, and data in the nth row after processing can be represented as

$$S_n(t_s, t_f) = \sum_{g=1}^{G} \sigma_g \sin c\left[B\left(t_f - \tau_m - \frac{2v}{c} t_s\right)\right] \tag{6}$$

where $\tau_m = 2(R_0 + \sqrt{\epsilon_w} R_w)/c$ stands for the fixed time delay caused by measurement environment, t_s denotes the time along slow time, B represents bandwidth, and $t_f = t_m - (n-1)T$ represents the time along fast time axis. The $(2v/c)t_s$ term arises from target moving. In ultrawideband radar system, the higher range resolution makes it easy for target to move across different range bins. If fast Fourier transform (FFT) is utilized to extract frequency information of each distance bin, the occurrence of range cell migration will induce the Doppler information to be distributed among multiple distance bins, resulting in broadening of the Doppler spectrogram. Consequently, it is necessary to compensate the range cell migration to obtain highly accurate distance components and high-resolution frequency components in the RD domain.

In this article, Keystone Transform [38] is employed to process data matrix $\mathbf{S}$ in order to eliminate the influence of range cell migration. Echo signal before Keystone Transform in (5) can be rewritten as

$$S_r(n, m) = \sum_{g=1}^{G} \sigma_g e^{-j2\pi f_m \tau_m} e^{-j2\pi f_m \frac{2vt_s}{c}}. \tag{7}$$

The keystone transform utilizes temporal scaling of the slow time dimension to transform data from various frequency points onto a single and common frequency point. The transformation is mathematically described by

$$t_s' = \frac{f_m t_s}{f_c} \tag{8}$$

where f_c is the predefined transformed frequency and $t_s{}'$ is the time along slow time after transformation. A higher frequency f_m corresponds to a longer time interval $t_s{}'$. In this case, the fast frequency/slow time domain will be stretched into an approximate trapezium shape, thus requiring resample along the slow time axis with a resample time of $t_s{}''$. In this article, f_c is set to 1.899 GHz, which is the center frequency of the frequency band. Besides, we achieve resampling by trimming the period length of each frequency point to the minimum period length (corresponding to 1.6 GHz), thereby transforming the fast frequency/slow time plane into a rectangular shape.

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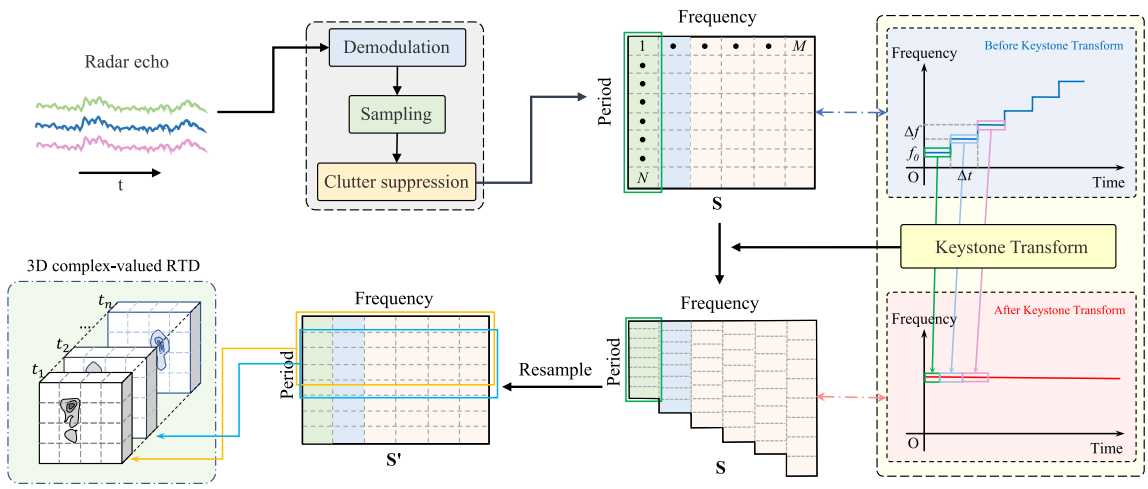

Fig. 2. Process flow of producing complex-valued RTD data from radar echoes.

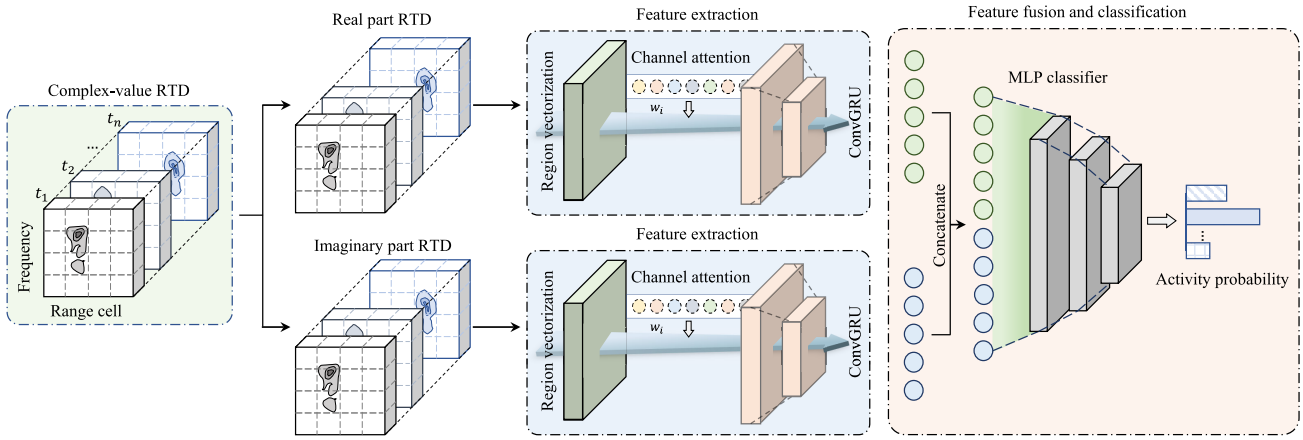

Fig. 3. Overall structure diagram of the proposed RV-ConvGRU network.

After keystone transform, (7) develops into

$$S_r'(n,m) = \sum_{g=1}^{G} \sigma_g e^{-j2\pi f_m \tau_m} e^{-j2\pi f_c \frac{2vt_s''}{c}}. \tag{9}$$

If IFFT is applied to the data within each period, the resulting fast time/slow time domain can be formulated as

$$S_n'\left(t_s'', t_f\right) = \sum_{g=1}^{G} \sigma_g \sin c\left[B\left(t_f - \tau_m\right)\right] e^{-j2\pi f_c \frac{2vt_s''}{c}}. \tag{10}$$

Comparing (6) with (10), it can be observed that after the keystone transform, the target is confined within the initial range cell.

To obtain RTD data, it is necessary to apply windowing to the slow time axis of the fast frequency/slow time domain after keystone transform. The desired short-time data segments are then selected by means of windowing to generate the RD map. By sliding the window and applying FFT on the data of each range cell, a series of RD maps can be produced into a 3-D data cube of RTD information. The flowchart for generating complex-valued RTD is shown in Fig. 2. In this article, we select the analyzed data segments through a window with length of 20 and a stride of 4.

III. HAR Method Based on RV-ConvGRU Network

To construct an effective HAR method using 3-D complex-valued data, a dual-channel ConvGRU based on region-vectorization (RV-ConvGRU) network is proposed, as illustrated in Figs. 3 and 4. To address the extensive computational and temporal costs associated with 3-D data, the 3-D data are partitioned and transformed into region feature vectors. By converting each region into a feature vector, local region information is preserved, and restructuring the feature vector into channels helps reduce the time complexity and spatial complexity of subsequent network. After that, channel attention mechanism [40] is introduced to improve the effectiveness of ConvGRU in focusing on advantageous channels after region-vectorization. Then, ConvGRU, which is capable of simultaneously obtaining time and spatial features, is applied to extract global features of RD domain and temporal features within RTD data. Furthermore, because the results from [34] and [41] demonstrate that dividing complex-valued data into real and imaginary parts leads to better outcomes, we design a dual-channel feature extractor to process the real and imaginary parts, and features of the real and imaginary parts are fused for classification.

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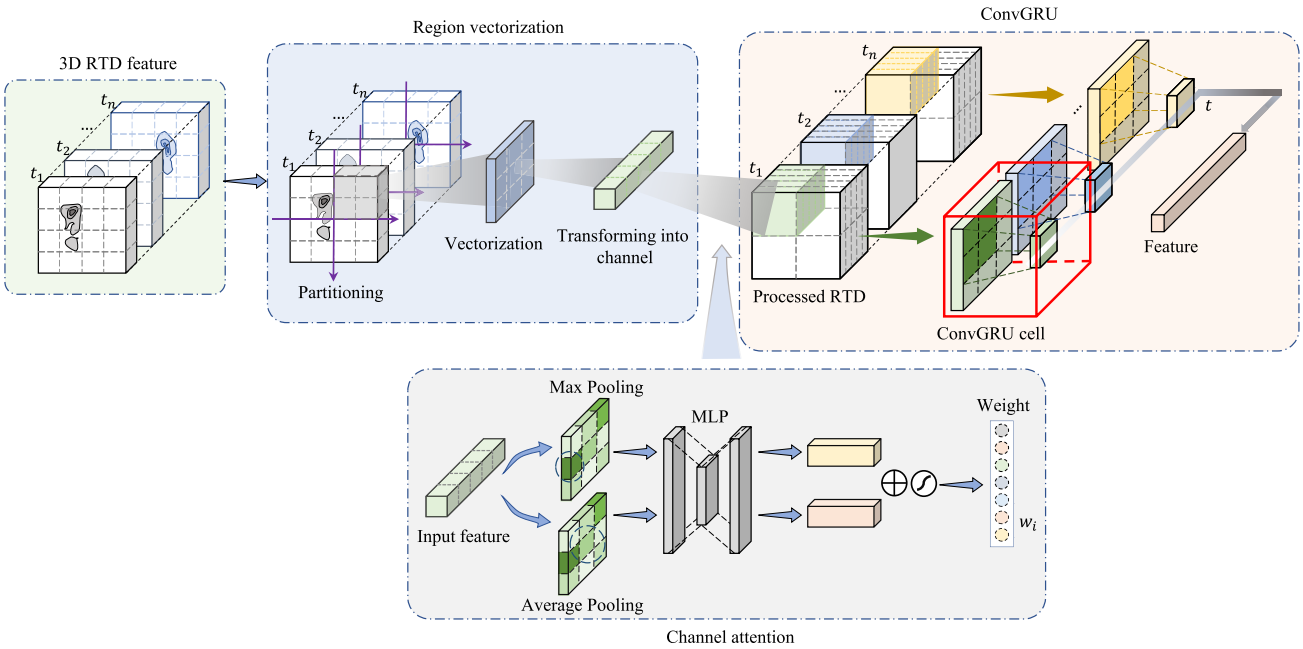

Fig. 4. Detailed information of the feature extraction part in RV-ConvGRU.

Generally, the processing procedure of proposed HAR method with RV-ConvGRU network is summarized as follows.

1) *Acquisition and Preprocessing of Radar Echoes:* Ultra-wideband SFCW radar is utilized to capture echoes of human activities behind wall. Subsequently, mixing, filtering, sampling, and clutter suppression are executed on echoes to obtain incipient data matrix $\mathbf{S}$.
2) *Generation of Complex-Valued RTD Data:* Keystone transform and resample are performed on $\mathbf{S}$ to obtain a new data matrix $\mathbf{S}'$. Then, designated periods of $\mathbf{S}'$ are selected to obtain RD map, and RD maps sequence is generated by continuously changing processing periods sliding along the slow time axis.
3) *Feature Extraction:* The real and imaginary parts of the complex-valued RTD data are separately input to the region-vectorization module to extract local features. Afterward, the channel attention module is used to increase selectivity on channels, and the ConvGRU is used to extract global and temporal features of the real and imaginary parts separately.
4) *Classification and Recognition:* The real and imaginary part features are concatenated and input to MLP for classification recognition, and the classification result is outputted.

A. Region-Vectorization Module

The region-vectorization module consists of three steps: partitioning, vectorization, and channels reorganization. Specifically, 3-D input data are divided into multiple nonoverlapping region blocks, each of which is vectorized into a region feature vector. Subsequently, the feature vectors are reconstructed into channels to form a new 3-D data. Compared to the original 3-D data, processed 3-D data has a reduced size in the RD domain, while also exhibiting an increased dimensionality along the channel axis. This effectively reduces the computational and spatial complexity of the subsequent network.

Assuming that the dimensionality of input RTD data is $H \times W \times L \times 1$ (height $\times$ width $\times$ length $\times$ channel), where H, W, and L correspond to the Doppler frequency, range, and the time axes, respectively, and the number of channel is 1. After partitioning, the resulting new 3-D data $\mathcal{RTD}$ can be formulated as

$$\mathcal{RTD} = \bigcup_{n_T=1}^{N_T} \bigcup_{n_R=1}^{N_R} \bigcup_{n_D=1}^{N_D} \mathcal{RTD}(n_T, n_R, n_D) \tag{11}$$

where N_T, N_R, and N_D represent the number of sliced blocks along the time, range, and Doppler frequency dimensions, while $\mathcal{RTD}(\cdot)$ denotes the region data after partitioning. In order to retain the temporal feature of sequence, partitioning is executed only on the RD plane, which means $N_T = L$.

Inspired by Qi et al. [42], in this article, 2DCNN is employed to transform the partitioned regions into feature vectors. Besides, maxpool function is used to reduce redundant features, the obtained vector is

$$\mathcal{V}_{\mathcal{RTD}(n_T,n_R,n_D)} = g_m\left(f\left(\bigcup_{i=1}^{I}(\mathcal{RTD}(n_T, n_R, n_D) \times w_i + b_i)\right), k\right) \tag{12}$$

where $g_m(\cdot)$ is max pooling function, k represents the kernel size, $f(\cdot)$ is activation function, w_i denotes weight of the ith convolution kernel, and b_i is bias. Afterward, $\mathcal{V}_{\mathcal{RTD}(n_T,n_R,n_D)}$ is rearranged as the channel of the respective region.

After being processed by the region-vectorization module, the dimensions of the 3-D data $\mathcal{RTD}$ are transformed into $N_D \times N_R \times L \times (\lfloor(H/N_D k)\rfloor \times \lfloor(W/N_R k)\rfloor \times I)$, where $\lfloor\cdot\rfloor$ is floor function. By comparing the sizes of the 3-D data before and after processing, it can be observed that the size of

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TABLE I
DIMENSION TRANSFORMATION OF CONVGRU BEFORE AND AFTER ADDING REGION-VECTORIZATION

Module	Step/Operation	Param setting	Output size
Region-vectorization +ConvGRU	Input	-	128×224×8×1
	Partitioning	$N_R = 16$, $N_D = 32$	8×7×8×1
	Conv2D	kernel_num=8	8×7×8×8
	Maxpool2D	kernel_size=2	4×3×8×8
	Channel reorganization	-	16×32×8×96
	ConvGRU	kernel_num=8	16×32×8
	Flatten	-	4096
ConvGRU	Input	-	128×224×8×1
	ConvGRU	kernel_num=8	128×224×8
	Flatten	-	229376

the RD domain decreases from $H \times W$ to $N_D \times N_R$, while the number of channels increases from 1 to $\lfloor (H/N_D k) \rfloor \times \lfloor (W/N_R k) \rfloor \times I$. The conversion of channelwise computations to interchannel computations results in a reduced output feature dimension after feature extraction, leading to a significant decrease in the complexity of the classifier. Additionally, the preservation of regional features through vectorization serves to fully utilize the information present in the RTD data.

In order to clearly illustrate the effect of the region-vectorization module, we provide numerical example in Table I. As shown in Table I, the input data has dimensions of 128 × 224 × 8 × 1 (range × Doppler × time × channel). After processed by the region-vectorization module, the dimension becomes 16 × 32 × 8 × 96, and then input into ConvGRU and flattened, resulting in a feature dimension of 4096. On the other hand, when the input data of size 128 × 224 × 8 × 1 is directly fed into ConvGRU and flattened, the resulting feature dimension is 229 376. It can be observed that region-vectorization module effectively reduces the output feature dimension, thereby reducing the complexity of the subsequent classifier.

B. Channel Attention Module

After region-vectorization module, the channel of the 3-D data is composed of regional feature vector. To enhance the network's capacity for focusing on effective features, a channel attention mechanism [40] is incorporated to weight the channels. Specifically, this mechanism dynamically assigns weights to each channel, which enables the network to place more emphasis on crucial information carried by relevant channels. The mathematical formulation for channel attention mechanism is provided as

$$M_c(\mathcal{RTD}) = \sigma(\mathrm{MLP}(g_a(\mathcal{RTD})) + \mathrm{MLP}(g_m(\mathcal{RTD}))) \quad (13)$$

where g_a represents the average pooling function, and MLP denotes multilayer perceptron. In this study, MLP is implemented through 2DCNN, with the number of perceptron layers set to 2 and the number of convolutional kernels equal to the number of input features.

C. ConvGRU

ConvGRU is a variant of the GRU, which aims to simultaneously extract temporal and spatial features of input data by replacing the fully connected calculation in the GRU with convolution operation. In the ConvGRU, the weights of each convolutional kernel are trained to adaptively capture the dynamic changes in the sequence. At each time step, the reset gate adjusts the retention of information, while the update gate controls the update of the hidden state from the previous time step. Specifically, at each time step, ConvGRU accepts the hidden state from the previous step and the input of the current step, and then calculates the hidden state and output of the current step through convolutional operations and gate control mechanisms, as shown in Fig. 5. The mathematical formulas for ConvGRU are

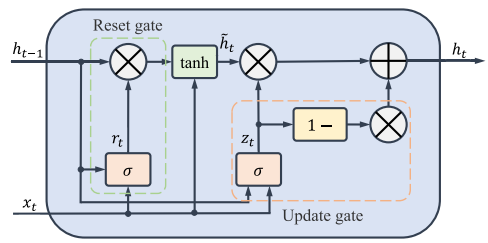

Fig. 5. Structure of ConvGRU cell.

$$z_t = \sigma(W_{xz} * x_t + W_{hz} * h_{t-1}) \quad (14)$$

$$r_t = \sigma(W_{xr} * x_t + W_{hr} * h_{t-1}) \quad (15)$$

$$\tilde{h}_t = \tanh\left(W_{xh} * x_t + r_t \circ \left(W_{h\tilde{h}} * h_{t-1}\right)\right) \quad (16)$$

$$h_t = (1 - z_t) \circ \tilde{h}_t + z_t \circ h_{t-1} \quad (17)$$

where x_t is the input at the current time t, h_{t-1} represents the state output at time $t-1$, W denotes the convolution kernel weight, z_t stands for the output of the update gate, r_t refers to the output of the reset gate, σ represents the sigmoid activation function, $\tilde{h}_t$ indicates the hidden state, and $\circ$ represents the Hadamard product.

D. Feature Fusion and Classification

The real and imaginary components of complex-valued RTD are first processed via the aforementioned region-vectorization module, channel attention module, and ConvGRU. The resulting features, denoted as F_{real} (real part) and F_{imag} (imaginary part), are further fused using concatenation operation to preserve the original sign property, instead of addition operation.

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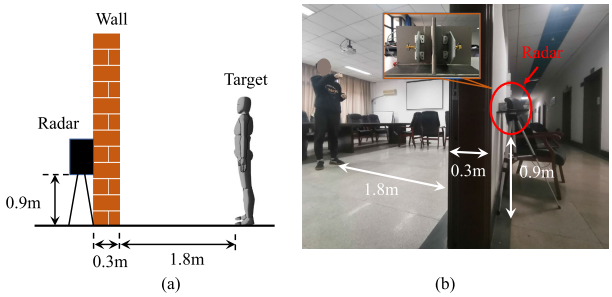

Fig. 6. Data collection scenario. (a) Schematic. (b) Actual scenario.

TABLE II
PARAMETERS OF THE ULTRAWIDEBAND SFCW RADAR

Parameter	Value
Starting frequency	1.6 GHz
Cut-off frequency	2.2 GHz
Stepped frequency	2 MHz
Pulse period	30.1 ms
Transmitting power	19 dBm
Range resolution	0.25 m

Concatenation operation can be expressed as

$$F_{\text{fusion}} = \left[F_{\text{real}}, F_{\text{imag}}\right]. \tag{18}$$

The fused feature vector is input to a classifier composed of fully connected layers. The computation of the first fully connected layer can be expressed as

$$F_{\text{fc}} = f(W \times F_{\text{fusion}} + B) \tag{19}$$

where the variables W and B represent the weight and bias of the fully connected layer. The calculation of the remaining fully connected layers in the classifier follows a similar pattern, differing only in the choice of activation function. Specifically, the last layer uses Softmax [43] as the activation function to obtain classification output, while the other layers use Rectified Linear Unit (ReLU) [44] as the activation function.

IV. Experimental Preparation and Implementation Details

A. Dataset

The main parameters of the ultrawideband SFCW radar employed in this article are presented in Table II. The experiment involves the acquisition of 11 daily activities, including boxing, drinking water, falling down, lifting leg, kicking, picking up, running, sitting down, standing up, walking, and waving. The wall is constructed with brick material, and its thickness is 0.3 m. The radar is placed behind the wall at a distance of 0.9 m away from the ground, with the target located 1.8 m away from the wall, and facing the radar, aligned with it on the same straight line, as shown in Fig. 6. During data collection, the target performs action in place. A total of ten volunteers, including eight males and two females with different heights and weights, are invited to participate in the data collection, resulting in the acquisition of 1650 segments of activity data (10 participants × 11 activities × 15 repetitions). The data are split into training and test sets with a ratio of 7:3. Due to the random shuffling and partitioning of all samples, there is a possibility that some samples from the same participant could be allocated to the training and test sets simultaneously. To the best of our knowledge, the diversity of the training set affects the model's generalization ability, while the test set is used to evaluate the model's generalization performance. Since random partitioning does not affect the diversity of the training set, we believe that the random partitioning method does not have an impact on the model's generalization performance.

TABLE III
DATASETS USED IN NETWORK EXPERIMENTS

Dataset name	Description
Dataset-I	Complex-valued 3D RTD dataset generated by using Keystone Transform.
Dataset-II	Complex-valued 3D RTD dataset generated without using Keystone Transform.
Dataset-III	Generated by taking amplitude value of Dataset-I.
Dataset-IV	Generated by taking amplitude value of Dataset-II.
Dataset-V	Three 2D amplitude datasets generated as [29].
Dataset-VI-TR	Complex-valued dataset of TR domain obtained by performing IFFT on the rows of the data matrix $\mathbf{S}$.
Dataset-VI-RD	Complex-valued dataset of RD domain obtained by performing FFT on the columns of Dataset-VI-TR.
Dataset-VI-TD	Complex-valued dataset of TD domain by performing short-time Fourier transform on the 150th column of $\mathbf{S}$.
Dataset-VII-TR	TR images dataset by taking amplitude value of Dataset-VI-TR.
Dataset-VII-RD	RD images dataset by taking amplitude value of Dataset-VI-RD.
Dataset-VII-TD	TD images dataset by taking amplitude value of Dataset-VI-TD.

To meet the requirements of subsequent network experiments, 11 datasets are generated using various methods, and detailed descriptions of these datasets are provided in Table III. It is worth noting that, all datasets involving distance are confined to a range of 0–5 m.

Fig. 7 illustrates the example of RTD amplitude images obtained with and without keystone transform. As can be seen from Fig. 7, the Doppler information of the limbs is severely weakened or even invisible due to the attenuation caused by wall penetration. Besides, the findings suggest that the two methods generate the almost same RD images when the target remains stationary within the fixed range unit (e.g., t_2). However, detectable differences between Doppler features of the two methods occur once the target undergoes cross-range unit motion (e.g., t_4, t_7). Notably, the Keystone Transform results in a more prominent reduction in Doppler bandwidth, indicating that it effectively mitigates the impact of range migration on the Doppler broadening effect (as confirmed by the image at t_7).

B. Evaluation Experimental Setting

To verify the effectiveness of proposed HAR method, the validation experiments are divided into three parts: ablation experiments, contrast with existing models, and performance research of the proposed method.

First, ablation experiments are performed to confirm the effectiveness of various modules, including keystone

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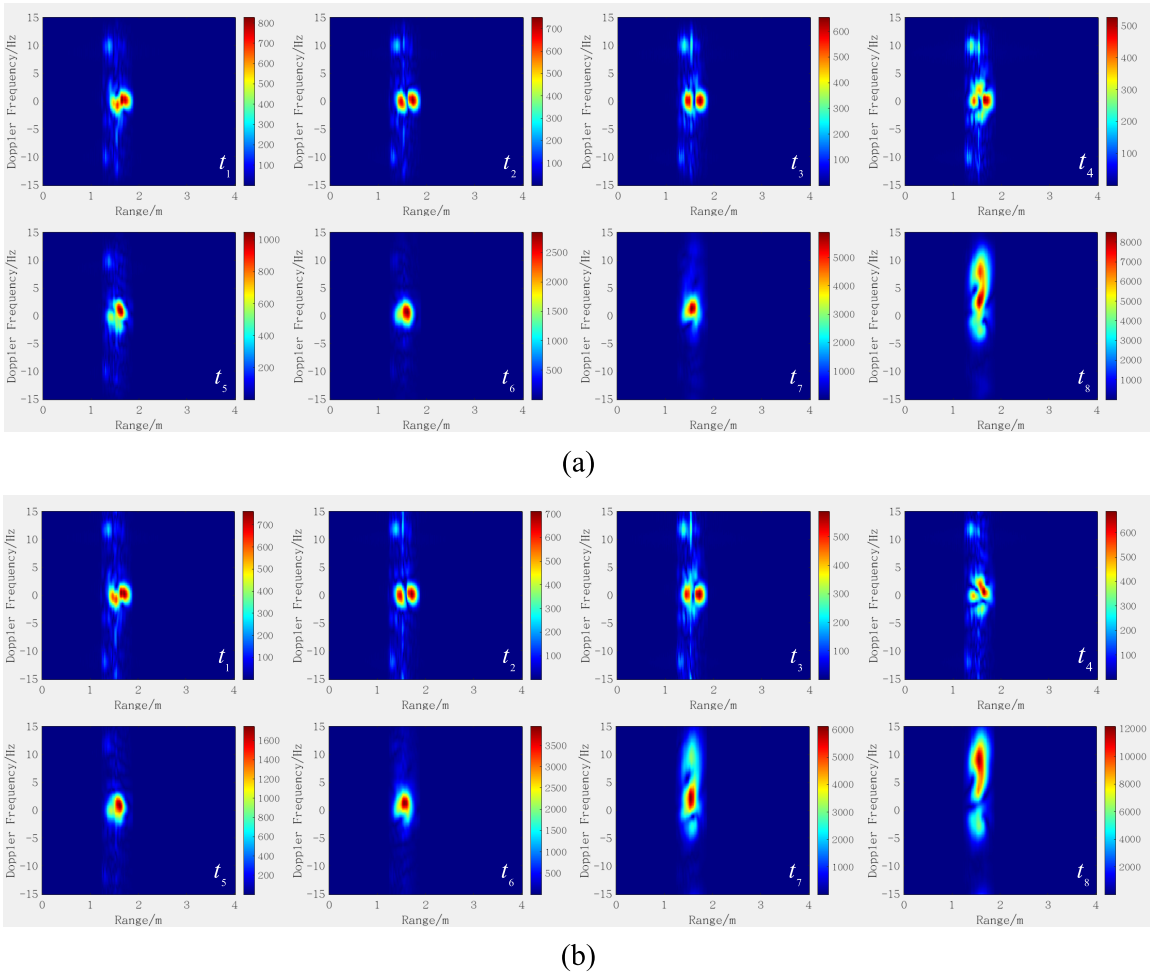

Fig. 7. Example of RTD amplitude image for lifting leg (a) with keystone transform (Dataset-III) and (b) without keystone transform (Dataset-IV).

transform, complex-valued information, region-vectorization module, and channel attention module.

Subsequently, comparisons with existing methods are executed. Specifically, for common 2-D features, classic networks such as VGG16 [45] and ResNet18 [43] are selected for comparison, and CNN-LSTM [46] is carried out for images with temporal information like RT and TD datasets. Then, contrasts are conducted with models based on 3-D data, including 3-D-OPEN [29] and Temporal 3DCNN [28]. Besides, neural networks employing complex-valued features, such as CV-VGG16 [32] and CV-CNN [47], are tested.

Finally, further experiments and discussions are conducted on important parameters involved in proposed method, such as the network layer parameters, input sequence length, and number of partitions, to explore the influence factors and performance limits of the proposed method.

The aforementioned experiments are conducted on a server equipped with NVIDIA GeForce RTX 4090 Ti GPU and PyTorch framework. In experiments, the training epochs are set to 150, the batch size is 64, the optimizer is Adam, and the initial learning rate is 0.001, which is decayed with a factor of 0.5 every 50 epochs. Furthermore, the negative log-likelihood loss is adopted for updating the network parameters, with its corresponding mathematical formula as

$$l(\theta) = -\sum_{x \in D_c} \log(P(x|\theta)) \tag{20}$$

where θ is network parameter, P denotes probability, and D_c stands for the class c in dataset. For the comparative experiments of existing models, the parameters are set according to the original paper. If the parameter is not specified in the original paper, it is set as described above.

C. Evaluation Metrics

Accuracy is the primary evaluation metric selected for this study. In order to mitigate the impact of the stochasticity from network training, experiments should be repeated many times, and the mean and standard deviation of accuracy are calculated as the results, which are calculated as

$$\mathrm{acc}_{\mathrm{mean}} = \frac{1}{N}\sum_{n=1}^{N} \frac{T_n}{T_n + F_n} \tag{21}$$

$$\mathrm{acc}_{\mathrm{std}} = \left(\frac{1}{N}\sum_{n=1}^{N}\left(\frac{T_n}{T_n + F_n} - \mathrm{acc}_{\mathrm{mean}}\right)^2\right)^{\frac{1}{2}} \tag{22}$$

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TABLE IV
RESULTS OF ABLATION EXPERIMENT

Dataset	Network	Acc (mean±std) (%)	FLOPs (G)	Params (M)
Dataset-I	ConvGRU	94.47±0.87	25.12	117.57
	ConvGRU with PCR	96.56±0.82	0.84	2.32
	ConvGRU with RVM	98.62±0.27	1.16	2.39
	ConvGRU with PCR and CAM	95.83±0.56	0.84	2.33
	The proposed method	98.95±0.30	1.16	2.39
Dataset-II	ConvGRU	94.47±1.22	25.12	117.57
	ConvGRU with PCR	94.22±0.54	0.84	2.32
	ConvGRU with RVM	96.60±0.30	1.16	2.39
	ConvGRU with CAM	94.50±0.56	0.84	2.33
	ConvGRU with RVM and CAM	96.56±0.49	1.16	2.39
Dataset-III	ConvGRU	95.19±1.38	12.96	58.80
	ConvGRU with PCR	96.16±0.57	0.82	1.27
	ConvGRU with RVM	96.32±0.92	0.61	1.20
	ConvGRU with CAM	95.19±0.86	0.82	1.32
	ConvGRU with RVM and CAM	95.76±0.93	0.61	1.27

where N denotes the repeat times of each experiment, which is set to five in this article. acc_{mean} and acc_{std}, respectively, refer to the mean and standard deviation of the accuracy. T_i and F_i represent the number of correct and incorrect judgments in the nth experiment.

Moreover, to compare comprehensively, floating point operations (FLOPs) and the number of parameters are utilized to measure the time and space complexity of the models.

V. EXPERIMENTAL RESULTS AND ANALYSIS

It is worth noting that the initial parameters used for both the ablation experiments and model comparison experiments are as follows. The length of the input RTD sequence is set to 8, corresponding to a duration of 1.44 s. In the region-vectorization module, the range and Doppler frequency dimensions are partitioned into 32 and 16 segments, respectively. The vectorization operation is performed using eight convolutional kernels, and a pooling kernel size of 2 is utilized. The ConvGRU network consisted of three layers, where the convolutional kernels in each layer are set to 32, 16, and 8. Additionally, the classifier comprises of three layers with 256, 64, and 11 neurons, respectively.

A. Ablation Experiments

To verify the effectiveness of each module in the proposed HAR method, ablation experiments are carried out on Dataset-I, Dataset-II, and Dataset-III. For simplicity, the partitioning and channel restructuring part in the region-vectorization module (without vectorization process), the complete region-vectorization module, and the channel attention module are recorded as PCR, RVM, and CAM, respectively, in the following discussions.

Table IV displays the results of the ablation experiments, where "ConvGRU" indicates that the dataset is directly fed into the ConvGRU network without passing through the RVM and CAM, and "Acc" represents accuracy. As shown in Table IV, for Dataset-I, partitioning and channel reconstruction operations in PCR significantly reduce the time and space complexity of network. This is because the feature maps output from the feature extraction module after PCR are substantially smaller, thus considerably reducing the subsequent computation of MLP, and due to the channelwise summation of convolution outputs, the computation is further reduced. After PCR, the accuracy of activity recognition is improved, as ConvGRU network is able to focus on effective global and temporal features. Following the addition of the RVM, the accuracy is further improved, since the vectorization module in the RVM could effectively preserve the validity of the local features, and the increase in nonlinearity also contributes to the network performance improvement. Furthermore, the CAM only plays a role after the RVM, as the information within a region exhibits high similarity, the direct use of CAM may even weaken the network performance. Compared to using ConvGRU alone, the method proposed in this article results in an average accuracy increase of 4.48% by adding RVM and CAM modules.

From the analysis of Dataset-II and Dataset-III, similar module-related conclusions can be drawn. However, it is worth noting that CAM only appears to have a positive effect in Dataset-I, while in other cases, it either produces no effect or even diminishes performance. One possible explanation is that the rich, accurate and clear feature information present in Dataset-I enhances the accuracy and representativeness of the associated feature vectors. As a result, the CAM is better able to focus on relevant features that are effective.

Furthermore, through a comparative analysis of the three datasets, it can be observed that the application of keystone transform and complex value in Dataset-I improve accuracy significantly, surpassing the accuracy of the other two datasets, despite the network complexity being higher than that of the amplitude dataset.

In order to elucidate the role of each module, confusion matrices of partial ablation experiments are provided in Fig. 8. The confusion matrix is plotted using the result from the experiment with the closest average accuracy out of the five experiments. The results show that the ConvGRU with PCR achieves relatively high recognition accuracy, as shown in Fig. 8(a). However, incorporating the CAM module directly may cause invalidation due to the similarity of features in

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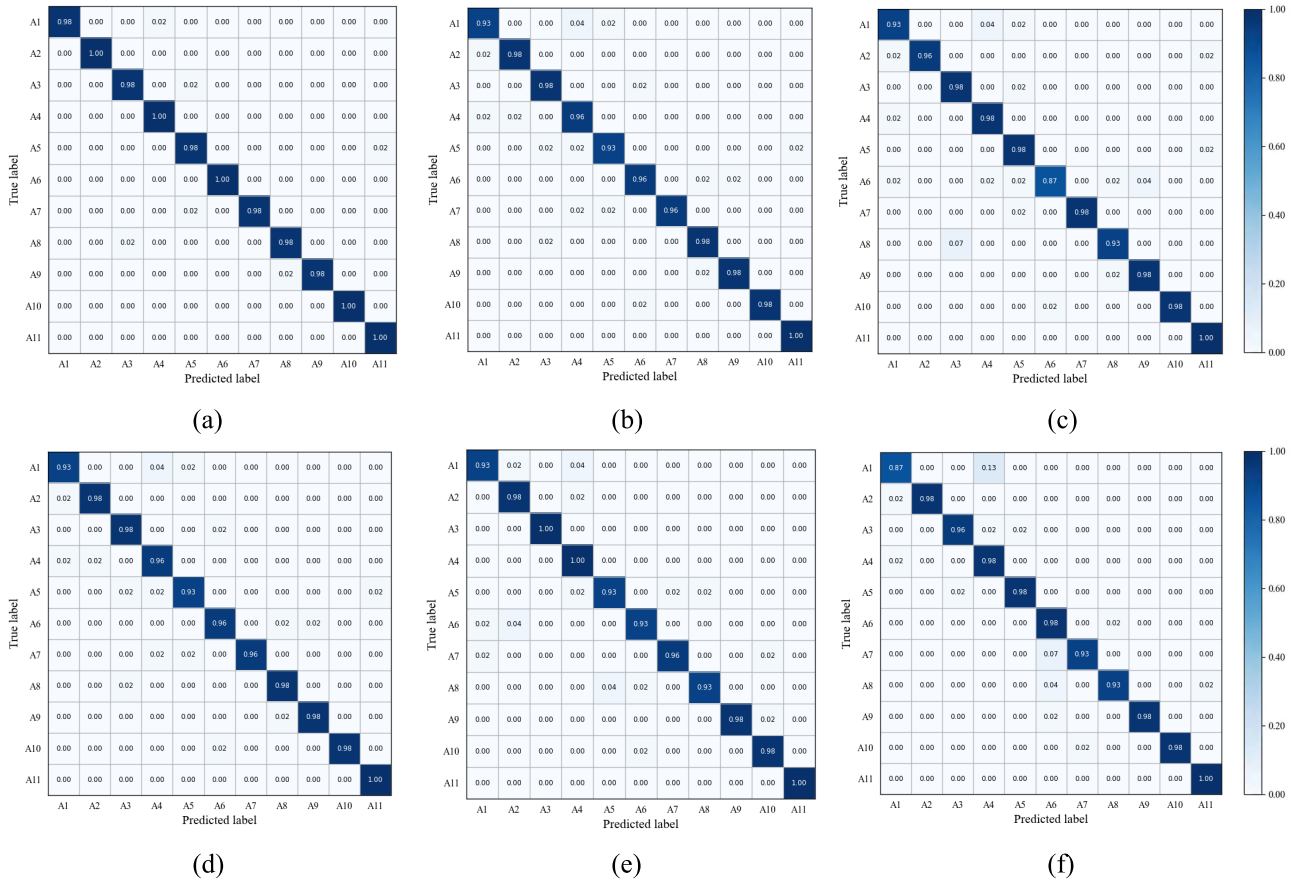

Fig. 8. Confusion matrices of ablation experimental results. (a) ConvGRU with PCR. (b) ConvGRU with CAM. (c) ConvGRU with RVM. (d) Proposed method. (e) ConvGRU with RVM and CAM on Dataset-II. (f) ConvGRU with RVM and CAM on Dataset-III. Additionally, actions A1–A11 represent boxing, drinking water, falling down, lifting leg, kicking, picking up, running, sitting down, standing up, walking, and waving, respectively.

the region, resulting in a decrease in recognition performance, displayed in Fig. 8(b). Introducing a vectorization operation after the addition of the CAM module (ConvGRU with RVM) effectively retains local features, leading to a further improvement in recognition accuracy for most actions. After adding the CAM module in this stage, as illustrated in Fig. 8(d), the attention focus of the model is shifted, which results in a change in the recognition accuracy of most actions. However, the attention mechanism improves the recognition accuracy of all actions to above 98%. As revealed by Fig. 8(d) and (f), the poor information content of amplitude features makes it easier to misclassify action A1 as A4, and the recognition accuracy of A7 and A8 is only 93%. Nonetheless, Fig. 8(e) shows that the 3-D complex-valued data provide more information, and it significantly improves the recognition accuracy for action A1. The dataset used in this study contains a rich amount of 3-D complex-valued data and optimizes feature representations using keystone transform, resulting in excellent recognition performance for all actions, as shown in Fig. 8(d).

In summary, the proposed method for HAR implements keystone transform to produce a complex-valued dataset containing abundant information and improved resolution of Doppler frequency. Within the proposed RV-ConvGRU, the RVM module effectively enhances recognition accuracy by retaining local features in relevant regions while reducing the time and space complexity of the network. Additionally, the ConvGRU module fully captures global and temporal features, while the CAM selectively emphasizes relevant local features to substantially improve the model's recognition performance.

B. Comparison Experiments With Other Models

Table V demonstrates the comparison results between the proposed model and other existing approaches. It should be noted that, in the experiments of VGG16 and ResNet18 based on 2-D images, the classifier of the models remained consistent with the proposed approach, and pretrained weights based on ImageNet are used, with an initial weight of 0.0001 for fine-tuning. Additionally, in the experiments of the Temporal 3DCNN model, as there lacks specific details regarding the number of convolutional kernels, the number of convolutional kernels is set to 32 and 16 for comparation. Moreover, in the experiment of CV-CNN, as the learning rate of 0.1 in the original paper [47] leads to gradient explosion while training on the dataset in this article, the learning rate is adjusted as the value of the proposed method. It is worth noting that, in order to ensure fairness in the comparison, the data length of the comparison dataset is also truncated to 1.44 s instead of being consistent with the original paper.

As seen in Table V, the accuracy of 2-D image-based networks is relatively low, with the highest recognition accuracy

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TABLE V
RESULTS OF COMPARATIVE EXPERIMENTS WITH EXISTING METHODS

Type	Dataset	Network	Acc (mean±std) (%)	FLOPs (G)	Params (M)
2D images	Dataset-VII-TR	VGG16 [45]	81.49±0.96	3.29	15.65
	Dataset-VII-RD		84.73±0.59	8.77	18.40
	Dataset-VII-TD		86.59±1.12	1.40	15.26
	Dataset-VII-TR	ResNet18 [43]	74.02±2.88	0.42	11.33
	Dataset-VII-RD		80.08±0.61	1.04	11.33
	Dataset-VII-TD		83.75±0.89	0.22	11.33
	Dataset-VII-TR	CNN-LSTM [46]	71.35±1.51	0.35	4.37
	Dataset-VII-TD		89.09±0.97	0.23	4.34
3D images	Dataset-V	3D-OPEN [29]	91.91±0.89	0.30	41.74
	Dataset-III	Temporal 3DCNN [28]	95.64±0.75	60.81	11.13
	Dataset-IV		92.65±2.04	60.81	11.13
Complex-valued 2D data	Dataset-VI-TR	CV-VGG16 [32]	91.15±2.45	13.11	30.36
	Dataset-VI-RD		94.79±0.41	34.95	33.12
	Dataset-VI-TD		85.45±1.30	5.49	29.97
	Dataset-VI-TR	CV-CNN [47]	76.73±3.50	0.006	0.13
	Dataset-VI-RD		93.49±0.35	0.018	0.35
	Dataset-VI-TD		81.29±1.21	0.003	0.05
Complex-valued 3D data	Dataste-I	The proposed method	98.95±0.30	1.16	2.39

being only 89.09%. Due to its deep network architecture, although VGG16 has higher time and spatial complexity, it performs well across all three 2-D images. ResNet18 has a high number of parameters, but its computational complexity is lower, with significant performance variations across the three kinds of images. For CNN-LSTM, its shallow network design results in lower time and spatial complexity, allowing for the highest recognition accuracy among TD images, while its performance is notably poor on TR images.

As shown in Table V, owing to the rich information in the 3-D images, the accuracy of methods based on 3-D images exceed 90%. Due to the fact that the feature extraction part of the 3-D-OPEN network is composed of three branches of EfficientNet, its time complexity is significantly lower than many other network structures, but its three-branch structure still results in a relatively high level of spatial complexity. For the Temporal 3DCNN network, the accuracy of dataset using keystone transform is significantly higher than that of the dataset without keystone transform, but both the time and space complexity of 3DCNN are relatively high.

For methods based on complex-valued 2-D data, the introduction of complex-valued representation enriches the characterization information, leading to generally higher recognition accuracy than methods based on 2-D images. However, in comparison to the VGG16 model based on 2-D images, which adopts pretraining methods, the recognition accuracy of the CV-VGG16 decreases on the Dataset-VI-TD, possibly due to the inability of the network to explore the suitable parameters during iteration. Furthermore, since complex-valued networks require constant combination and decomposition operations between the real and imaginary parts at different layers, the time complexity is significantly higher than that of the VGG16 model based on real-valued networks. Although the recognition accuracy of CV-CNN is inferior than that of CV-VGG16, its time and spatial complexity are significantly lower.

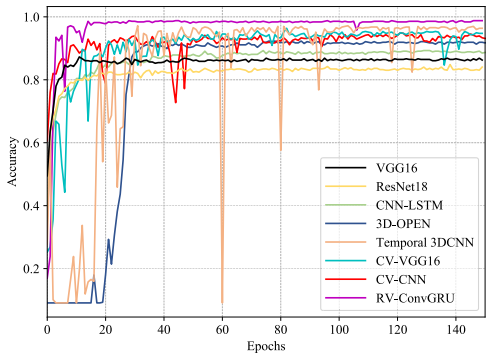

Fig. 9. Performance of different models on the test set during the training process.

In order to further demonstrate the model's training convergence, we plot accuracy curve on the test set during training in Fig. 9. Fig. 9 is generated by selecting the result from an experiment with the closest average accuracy out of the five experiments. To ensure the readability of the image, the highest recognition rate for each network is chosen, and the data for the first 150 epochs of training are retained. The results show that the methods based on 2-D images converge faster, while those based on 3-D images converge slower. The method proposed in this study exhibits a convergence speed somewhere between the two. Furthermore, our proposed method exhibits less fluctuation after initial convergence.

To sum up, the proposed HAR method outperforms existing models based on 2-D images, 3-D images, and complex-valued 2-D data in terms of recognition accuracy. Notably, our method has a comparable time complexity to that of the 2-D image-based methods, and a lower spatial complexity than the 2-D image-based methods.

C. Research Experiments on Method Performance

To investigate the performance bounds of the method and the influence of its parameters, experiments are conducted on the important tunable parameters of the method.

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TABLE VI
EXPERIMENTAL RESULTS OF CHANGING NETWORK PARAMETERS

Parameter setting	Acc (mean±std) (%)	FLOPs (G)	Params (M)
8-32-16-8	98.95±0.30	1.16	2.39
8-16-16-8	98.14±0.20	0.59	2.25
8-32-32-8	98.67±0.42	1.47	2.47
8-64-32-16	98.82±0.39	3.16	4.98
16-16-16-8	98.18±0.49	0.97	2.35
16-32-16-8	98.91±0.40	1.22	2.41
8-16-16	98.06±0.20	0.55	4.34
8-32-16	98.79±0.40	1.12	4.48
8-32-32	98.08±0.10	1.40	8.74
8-32-16-8-4	98.59±0.61	1.16	1.34
8-32-32-16-8	98.91±0.37	1.16	2.50
8-32-16-16-8	98.90±0.45	1.27	2.42

1) Impact of Network Parameter: Table VI presents the results of adjusting the vectorization parameter and the number of convolutional kernels in ConvGRU. The first number in the parameter settings denotes the number of features generated from the region-vectorization module, while the following numbers are the quantity of convolutional kernels in each layer of ConvGRU.

The results reveal that reducing the number of convolutional kernels in the final layer of ConvGRU can effectively keep the spatial complexity of the model at a relatively low level. This is because the parameters of the final layer of ConvGRU significantly influence the number of output features, and subsequently determine the number of parameters in subsequent classifier. Moreover, when the number of convolutional kernels in the final layer of ConvGRU is set to 32, numerous output features, including both effective and redundant ones, are produced, which may be harmful to subsequent classifier. However, when the number of convolutional kernels in the first layer of ConvGRU is reduced (i.e., set to 16), the accuracy of the network tends to decrease. The reason for this is that the number of convolutional kernels in the first layer of ConvGRU directly affects the breadth of the feature extraction from the image. A smaller number of convolutional kernels will lead to fewer features obtained from the image, resulting in lower classification performance. Under the adjustment of other parameters, the recognition accuracy of the HAR method remains consistently high. It is concluded that the recognition accuracy of this HAR method is marginally affected by network parameters.

2) Impact of Time Length: The performance of a model in terms of input time length reflects its real-time processing capability. In this article, by adjusting the input length (L) of 3-D complex-valued RTD data, we investigate the real-time processing performance of the network, as illustrated in Fig. 10. The results reveal a positive correlation between the input sequence length and the richness of input information, manifested as the recognition accuracy increasing with the length of the sequence. Especially, the accuracy of the network shows a significant boost when the quantity of inputs changed from 1 to 2. However, when the quantity changed from 8 to 9, the average accuracy of the network declines from 98.95% to 98.87%. It is worth noting that, despite the decrease in average accuracy, the maximum accuracy still remains high. This phenomenon of lower average accuracy could be attributed to the randomness of the training process.

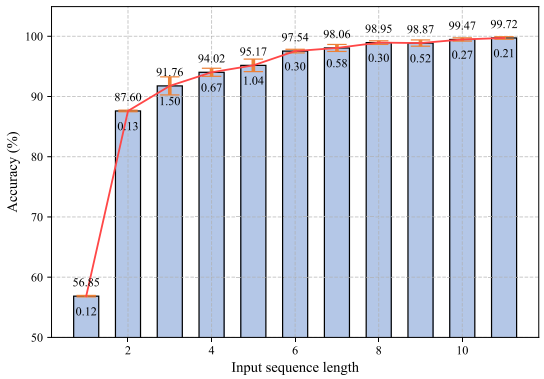

Fig. 10. Results of different input sequence lengths.

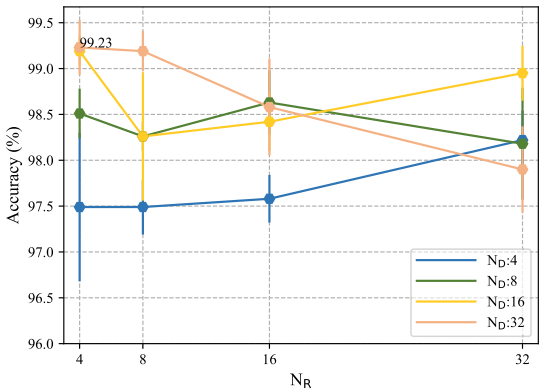

Fig. 11. Experimental results on changing the number of blocks.

By analyzing Fig. 10 and Table V, it can be inferred that the proposed method achieves a recognition accuracy equivalent to that of 2-D image-based methods with sequence length of 2 (0.72 s). Furthermore, our method achieves a recognition accuracy comparable to that of 3-D image-based methods by using input sequences with length of 4, lasting 0.96 s, and displays superior real-time performance.

3) Impact of the Number of Blocks: In order to explore the influence of region-vectorization module on the network, a thorough investigation of block partition refinement is conducted. The results of this study are presented in Fig. 11. Our findings indicate that increased refinement of the range axis partitioning, as measured by an increase in N_R, leads to a decrease in recognition accuracy. Conversely, an upward trend in recognition accuracy is observed when partitioning of the Doppler frequency axis is refined and N_R is small, but that trend becomes negative as N_R increases. These trends are attributed to the fact that target features in the RD plane tend to gather sparsely within a specific small-scale range. As such, a small number of range axis blocks facilitates feature aggregation and efficient feature extraction. In contrast, because Doppler features are more widely distributed on the Doppler axis with more densely distributed differences, finer partitioning of the Doppler axis is necessary when N_R is small in order to achieve higher recognition accuracy.

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In summary, the proposed method is sensitive to region partitioning, where appropriate partitioning may attain a higher level of accuracy. Specifically, by setting N_R to 4 and N_D to 32, the recognition accuracy reaches 99.23%.

VI. Conclusion

In this article, a novel method to recognize human behavior behind wall by utilizing 3-D complex-valued RTD data and RV-ConvGRU is presented. The 3-D complex-valued data offer a complete and comprehensive feature representation of human behavior under through-wall condition, enhanced by the use of the Keystone Transform to optimize the Doppler features. To address the computational challenges posed by the 3-D complex-valued data, we propose a low-complexity network, RV-ConvGRU, that includes a region-vectorization module to reduce output feature size and decline computations. Our proposed method effectively extracts local, global, and temporal features of the RTD and achieves better recognition accuracy as demonstrated by extensive experiments and ablation studies. The proposed method is compared against existing methods based on 2-D images, 3-D images, and 2-D complex-valued data to demonstrate its effectiveness. Additionally, we conduct a thorough study of important parameters to explore the performance limits of the network, revealing that average recognition accuracy of 99.23% can be achieved with an input sequence length of 1.44 s.

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